535/1
PHYSICS
Paper 1
July - August 2025
2½ hours



KAMSSA JOINT MOCK EXAMINATIONS

Uganda Certificate of Education

PHYSICS

Theory

Paper 1

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

- *This paper consists of two sections; **A** and **B** It has **seven** examination items..*
- *Section **A** has **three** compulsory items.*
- *Section **B** has **two** parts; **I** and **II**. Answer one item from each part.*
- *Answer **five** items in all.*
- *Any additional item(s) answered will not be scored.*
- ***All** answers must be written in the booklets provided.*

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SECTION A

*Answer **all** the items in this section*

ITEM 1.

A physics teacher presented an activity to senior three students in your school. He wanted them to explain the following occurrences.

- During hot days, the sky appears to have certain colors during mid- day, which are different from those in the morning.
- Similarly, the tarmacked roads seem to have an illusion like a pool of water when observed obliquely from a distance.

The senior three students are unable to do the classroom activity due to lack of knowledge and have approached you for help.

Task:

Assist the senior three students in doing the classroom activity.

Hint

Colour of light	Frequency (Hz)
White	7.9×10^{14}
Blue	6.100×10^{14}

Velocity of electromagnetic radiation=3.0×10^{8} ms^{-1}

ITEM 2

A patient has been taken by parents to a referral hospital, complaining of several pains in the head. A radiotherapist has admitted the patient in an isolated room and decided to examine the patient for a brain tumor. In the process of the examination, a radioactive isotope has been injected into the patient to be used in diagnosis of the tumor.

The activity of the radioactive isotope from the brain is detected and recorded as shown in table 1.

Activity (counts per minute]	250	140	76	38	25	18
Time (minutes]	0	5	10	15	20	25

Whenever the parents of the patient went to check on him in the isolated room, the radiotherapist tells them to wait for the right time (when the activity is equal to the background count rate of 50).

The doctor has also informed the parents that if the right time for them to check on the patient comes, they must observe precautions in the isolated room.

The parents have demanded to know the period they should wait before they can check on their patient but no one answers them. They have also been left wondering what the precautions are and why they should observe them.

You have heard the doctor talking to the parents and they have contacted you to address their concerns.

Task

(a) (i) Use the data to determine how long the parents should wait before they are allowed to check on their patient.

(ii) Explain to the parents why they need to wait for that time before they enter the isolated room.

(b) (i) State the precautions the relatives have to observe inside the isolated room.

(ii) Explain why the parents should observe the precautions while inside the isolated room.

ITEM 3

Some regions in northern Uganda are affected by the changes in seasons that sometimes may lead to famine. Accordingly, the people in that region have been demanding for solutions from the government to ensure that famine does not occur any more.

The government is planning to launch a second satellite into the lower Earth orbit through the support of the international space station (ISS). A number of people have criticized the move claiming that the satellite will not solve the problem of famine in that region.

Task

(a) help the public to understand the occurrence of seasons.

(b) Make the people understand and appreciate the reason why the government of Uganda needs to send these satellites in space.

SECTION B

Part I

Answer one item from this part

ITEM 4

A mountain climber of mass 80 kg is provided with two pairs of shoes where one has sharp spikes and the other has flat soles. The climber decides to follow the path which is slippery and had difficulties in climbing. The climber records the barometric reading at the bottom and the top of the mountain as 76cmHg and 72cmHg respectively. On reaching the top of the mountain, the climber felt very tired, had nose bleeding and had difficulty in getting the food ready. The mountain climber wondered the cause of all these problems.

Hint:

Density of air $=1.20 kgm^{-3}$

Density of mercury $=1.36 \times 10^{4}\ kgm^{-3}$

Acceleration due to gravity $= 10 ms^{-2}$

Task: As a physics learner help the mountain climber to understand;

(a) why there was difficulty in climbing the mountain.

(b) The cause of;

(i) being very tired

(ii) nose bleeding

(c) why there was difficulty in getting the food ready.

ITEM 5

The operator of a hot air balloon wishes to transport a boat engine from the landing site to the next island. The operator has a mass of 75 kg and the co-operator has a mass of 80 kg and wishes to use a hot air balloon having a fabric of mass 10kg. The balloon has a volume of 200 m^3 and has to be pumped with hot hydrogen gas of density 0.9 kgm^{-3} so as it can fly in air of density 1.25 kgm^{-3}. The inflow and outflow temperature of hydrogen is 100^0C and 10^0C respectively. However, the balloon cannot move if the mass of the engine is more than 79 kg. The co-operator demands to know why the hydrogen to be used must be hot, why the balloon is able to move in air yet it is a solid and how the balloon is able to land back from air and the quantity of heat needed for the hydrogen.

Task

As a learner of physics;

(a) Determine whether the balloon will fly or not.

(b) Address the demands of the co-operator

Hint

The specific heat capacity of hydrogen $=14300\ Jkg^{-1}K^{-1}$.

Part II
Answer one item from this part

ITEM 6

The welders in a certain workshop are troubled with their tools being shocked by electricity from a generator of 240V. When they visited a technician, they were advised to wind a copper wire to 3000 turns around a rectangular soft iron frame on the receiving part so as to output 120V suitable for their work shop operations fixed in a box. However, the welders seem bothered of how this will be helpful.

Task

As a learner of physics,

(a) Explain to the welders how the above design will produce power corresponding to their consumption.

(b) Help the welder to determine the number of turns to be wound on the output part of the device.

(c) Comment on how efficient the device is if the ratio of current output to current input is 1.6.

(d) Advise the welders on how to improve the efficiency of the device.

ITEM 7

In one of the secondary schools in a remote village, the students lack enough physics apparatus. The government decided to donate apparatus, including many moving coil galvanometers through a local supplier. However, there was still a scarcity of ammeters in the school. Each galvanometer has a resistance of 4 ohms and a full-scale deflection of 25 mA. The Head teacher wants to return some of the galvanometers back to the supplier in exchange for ammeters. The Head teacher does not know the use and mode of operation of these galvanometers. However, the Physics teacher advises the Head teacher that these galvanometers can be modified to work as ammeters and can read up to a current of 1A. He added that this project can be entrusted to the senior four class where you are the physics class coordinator.

Task

Advise the Head teacher on how you are going to perform this task and explain to him how the galvanometers work.

END

END

KAMSSA UCE PHYSICS
PAPER 1 SCORING GUIDE 2025
SECTION A

ITEM 1.

A physics teacher presented an activity to senior three students in your school. He wanted them to explain the following occurrences.

- **During hot days, the sky appears to have certain colors during mid- day, which are different from those in the morning.**
- **Similarly, the tarmacked roads seem to have an illusion like a pool of water when observed obliquely from a distance.**

The senior three students are unable to do the classroom activity due to lack of knowledge and have approached you for help.

Task:

Assist the senior three students in doing the classroom activity.

Hint

Colour of light	Frequency (Hz)
White	7.9×10^{14}
Blue	6.100×10^{14}

Velocity of electromagnetic radiation=3.0×10^{8} ms^{-1}

For white colour

$v = f\lambda$

$3.0 \times 10^{8} = 7.9 \times 10^{14}\lambda$

$\lambda = 3.797 \times 10^{-7}$m

For blue light

$v = f\lambda$

$3.0 \times 10^{8} = 6.100 \times 10^{14}\lambda$

$\lambda = 4.918 \times 10^{-7}$m

MC_1 - 03 (4-5)
- 02 (3-2)
- 01 (1)
- 0 (0)

Note: From the calculation, the frequencies in the item were altered and therefore they could not give the correct values of wave length for blue and white light.

(ii) The Phenomena soon is a mirage

A mirage appears on a tarmacked road on a hot day.

On a hot day, the air near the road is warm and less dense than that higher up.

Rays of light from the sun are continuously and progressively refracted away from the normal as light moves from more to less dense air. At some point, the angle of incidence is greater than the critical angle and total internal reflection occurs. The rays of light from the sky go back and an observer sees the image of the blue sky as a pool of water on the road surface. The…….. of the pool of water is known as mirage.

IR_1 - 03 (7-10)
- 02 (3-6)
- 01 (1-2)
- 0 (0)

TOTAL MAXIMUM SCORE = 06 SCORES

ITEM 2

A patient has been taken by parents to a referral hospital, complaining of several pains in the head. A radiotherapist has admitted the patient in an isolated room and decided to examine the patient for a brain tumor. In the process of the examination, a radioactive isotope has been injected into the patient to be used in diagnosis of the tumor.

The activity of the radioactive isotope from the brain is detected and recorded as shown in table 1.

Activity (counts per minute]	**250**	**140**	**76**	**38**	**25**	**18**
Time (minutes]	**0**	**5**	**10**	**15**	**20**	**25**

Whenever the parents of the patient went to check on him in the isolated room, the radiotherapist tells them to wait for the right time (when the activity is equal to the background count rate of 50).

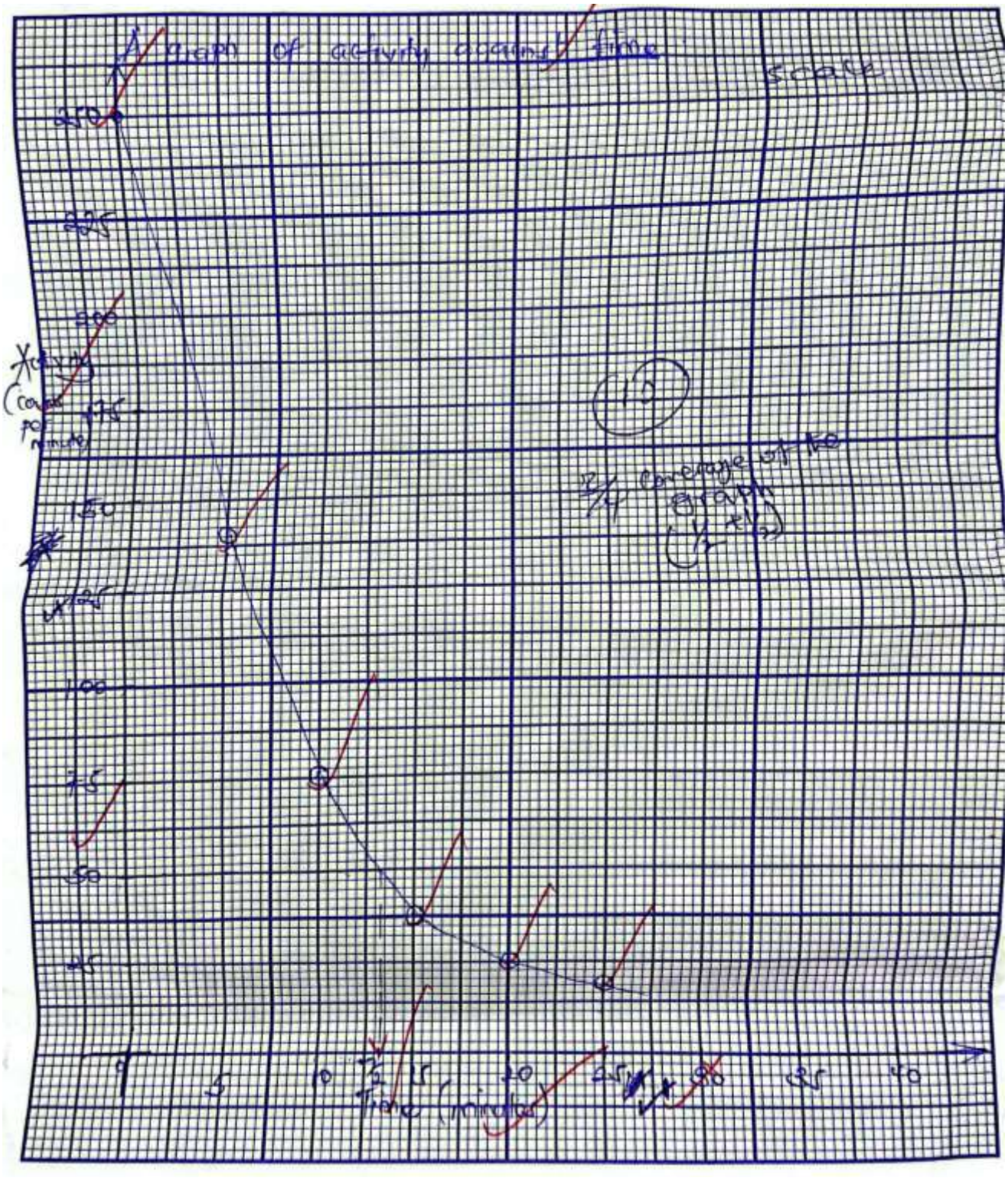
The doctor has also informed the parents that if the right time for them to check on the patient comes, they must observe precautions in the isolated room.

The parents have demanded to know the period they should wait before they can check on their patient but no one answers them. They have also been left wondering what the precautions are and why they should observe them.

You have heard the doctor talking to the parents and they have contacted you to address their concerns.

Task

(a) (i) Use the data to determine how long the parents should wait before they are allowed to check on their patient.



From the graph, the parents should wait for 13.25minutes

(ii) Explain to the parents why they need to wait for that time before they enter the isolated room.

Before that time of 13.25minutes, the radiations from the radioactive materials are highly dangerous and may cause harm to parents.

However, after that time, the materials will not cause any danger to the lives of the parents since the levels of radiation would have reduced below the background radiation.

(b) (i) State the precautions the relatives have to observe inside the isolated room.

- The parents should not eat or drink while inside the isolation room
- They should put on personal protective gears e.g gloves, boots, lead coats etc
- They should avoid direct contact with the parents

- They should reduce the time of expose inside the room
- In case they have wounds, they must cover them with thick wool

(ii) Explain why the parents should observe the precautions while inside the isolated room.

- Cause cancer
- Cause sterility of infertility
- Cause skin burn/skin irritation
- Cause genetic mutation
- Cause blindness

IR_2 - 03 (15-23)
- 02 (8-14)
- 01 (1-7)
- 0(0)

TOTAL MAXIMUM SCORE = 03 SCORES

ITEM 3

Some regions in northern Uganda are affected by the changes in seasons that sometimes may lead to famine. Accordingly, the people in that region have been demanding for solutions from the government to ensure that famine does not occur any more.

The government is planning to launch a second satellite into the lower Earth orbit through the support of the international space station (ISS). A number of people have criticized the move claiming that the satellite will not solve the problem of famine in that region.

Task

(a) help the public to understand the occurrence of seasons.

- Seasons occur due to the tilt of the earth with respect to the orbit and revolution around the sun that occurs throughout the year.
- The part of the hemisphere of the earth that is close to the sun receives more sunlight energy and hence experiences longer day time which gives a dry season or summer.
- The other parts of the hemisphere that face away from the sun receive less sunlight energy and hence experience a shorter day time when gives wet season or winter.
- During times when day and night time are equal, ie. When the sun over heads the equator, spring and Autum are experienced.

(b) Make the people understand and appreciate the reason why the government of Uganda needs to send these satellites in space.

- The satellite to be launched is known as weather satellite.
- This satellite has instruments like cameras, pyrometers, barometers etc for collecting data.
- After fine data has been collected, it is sent to a ground station in form of radio waves.
- The data is now processed, and analysed by meteorologists who make accurate predictions of the weather in the nearby future.

Accept other uses of satellites like

- Surveillance
- Communication
- Research
- Navigation
- Observation

- Making phone calls
- Live broad cast

Any four

IR_3 - 03 (7-10)
- 02 (3-6)
- 01 (1-2)
- 0(0)

TOTAL MAXIMUM SCORE = 03 SCORES

SECTION B

Part I

Answer one item from this part

ITEM 4

A mountain climber of mass 80 kg is provided with two pairs of shoes where one has sharp spikes and the other has flat soles. The climber decides to follow the path which is slippery and had difficulties in climbing. The climber records the barometric reading at the bottom and the top of the mountain as 76cmHg and 72cmHg respectively. On reaching the top of the mountain, the climber felt very tired, had nose bleeding and had difficulty in getting the food ready. The mountain climber wondered the cause of all these problems.

Hint:

Density of air =1.20kgm^{-3}

Density of mercury =1.36x10^4 kgm^{-3}

Acceleration due to gravity = 10ms^{-2}

Task: As a physics learner help the mountain climber to understand;

(a) Why there was difficulty in climbing the mountain.

The climber used shoes which were flat shoes and used a slippery road. There was less friction on the road and hence less grip. Therefore, climbing was supposed to be difficult.

(b) The cause of;

(i) being very tired

P.E = mgh

P.E = 80 x 10 x h

Pressure of air = Pressure difference

$p_a hg = p_m h_2 g - p_m h_1 g$

$1.2h = 1.36 \text{ x } 104\ (0.76 - 0.72)$

$h = 453.33m$

$P.E = 80 \text{ x } 10 \text{ x } 453.33$

$= 362664J$

The peson feels very tired because he/she spends 362664J of energy to climb as he/she is overcoming gravitational force.

MC_4 - 03 (7-9)
- 02 (3-6)
- 01 (1-2)
- (0)

IR_4 - 03 (7-11)
- 02 (3-6)

- 01 (1-2)
- (0)

(ii) nose bleeding

The blood capillaries in the nose burst due to the pressure difference of 4cmHg. As the person climbs, pressure reduces with altitude.

(c) why there was difficulty in getting the food ready.

At the top of mountain, atmospheric pressure is lower than at the bottom. The liquid boils when atmospheric pressure is equal to the saturated vapour pressure. Since pressure is lower at the top of mountain, even boiling point will be lower. Therefore, the food will boil at a lower temperature which makes it take long to get ready.

The person should use pressure cooker to make food get ready quickly.

TOTAL MAXIMUM SCORE = 6 SCORES

ITEM 5

The operator of a hot air balloon wishes to transport a boat engine from the landing site to the next island. The operator has a mass of 75 kg and the co-operator has a mass of 80 kg and wishes to use a hot air balloon having a fabric of mass 10kg. The balloon has a volume of 200 m^3 and has to be pumped with hot hydrogen gas of density 0.9 kgm^{-3} so as it can fly in air of density 1.25 kgm^{-3}. The inflow and outflow temperature of hydrogen is 100^0C and 10^0C respectively. However, the balloon cannot move if the mass of the engine is more than 79 kg. The co-operator demands to know why the hydrogen to be used must be hot, why the balloon is able to move in air yet it is a solid and how the balloon is able to land back from air and the quantity of heat needed for the hydrogen.

Task

As a learner of physics;

(a) Determine whether the balloon will fly or not.

Upthrust, u $= \rho_a vg$

$= 1.25\ x\ 200\ x\ 10$

$= 2500N$

$Wh = \rho_h vg$

$= 0.09\ x\ 200\ x\ 10$

$= 180N$

$Wb = mg$

$= 10\ x\ 10$

$= 100N$

$Wp = mg$

$= (75 + 80)\ x\ 10$

$= 1550N$

$At\ equilibrium\ W_h + W_b + W_p + W = U$

$180 + 100 + 1550 + W = 2500$

$W = 670N$

$W = mg$

$670 = MNO$

$M = 67kg$

Since mass of engine 67kg is less than 79kg, then the ballon will fly.

MC_4 - 03 (15-20)
- 02 (5-14)
- 01 (1-4)
- (0)

IR_4 - 03 (6-9)
- 02 (3-5)
- 01 (1-2)
- (0)

(b) Address the demands of the co-operator

Hint

The specific heat capacity of hydrogen =14300 $Jkg^{-1}K^{-1}$.

- Hot hydrogen is less dense than cold one. When pumped into the ballon, it is less dense than air enabling it to rise in space.
- The ballon is able to move in air because it is hollow, contains hot hydrogen and has a big volume making it less dense than air.
- When the ballon is required to land, cold hydrogen is pumped in making it more dense than air hence fall……

$Q = mc\Delta\theta$

$= \rho vc\Delta\theta$

$= 0.9 \times 200 \times 14300 \times (100 - 10)$

$= 231{,}660{,}000J$

Mass of $H_2 = 180kg$

The quantity of heat supplied to hydrogen gas to make it move the ballon is 231,660,000J

TOTAL MAXIMUM SCORE = 6SCORES

Part II

Answer one item from this part

ITEM 6

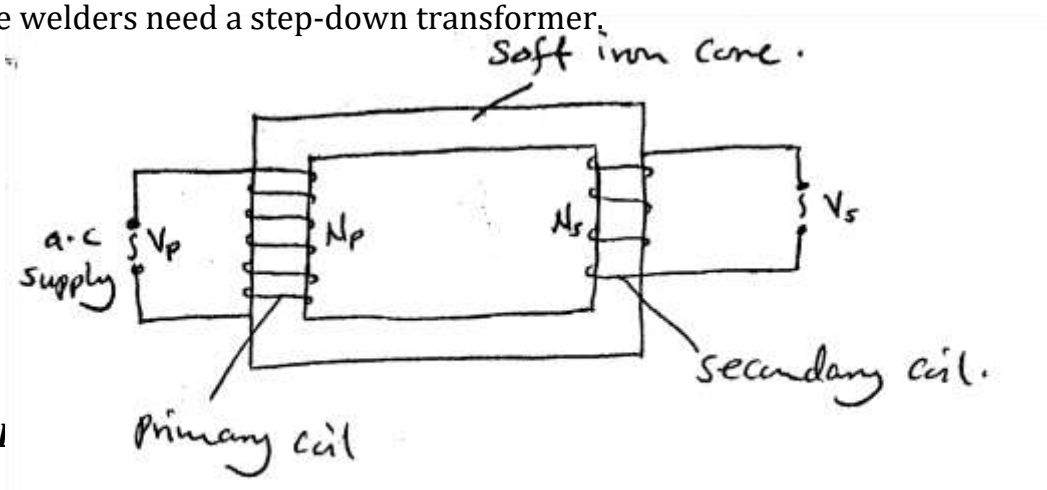
The welders in a certain workshop are troubled with their tools being shocked by electricity from a generator of 240V. When they visited a technician, they were advised to wind a copper wire to 3000 turns around a rectangular soft iron frame on the receiving part so as to output 120V suitable for their work shop operations fixed in a box. However, the welders seem bothered of how this will be helpful.

Task

As a learner of physics,

(a) Explain to the welders how the above design will produce power corresponding to their consumption.

The welders need a step-down transformer.



When alternate voltage (V_p) is connected to the primary coil, it produces alternating current (J_p) which causes alternating magnetic flux in the primary coil. This flux links the secondary coil leading to voltage (V_s) in the secondary coil. Since number of turns (N_p) in primary coil is greater than number of turns in secondary coil (N_s), then the transformer is called step down transformer.

(b) Help the welder to determine the number of turns to be wound on the output part of the device.

$$\frac{N_s}{N_p} = \frac{V_s}{V_p}$$

$$N_s = \frac{V_s N_p}{V_p}$$

$$= \frac{120 \text{ x } 3000}{240}$$

$$N_s = 1500 \text{ turns}$$

Therefore, the number of turns to be wound in the output part is 1500 turns.

MC - 03 (7-10)
- 02 (3-6)
- 01 (1-2)
- 0 (0)

(c) Comment on how efficient the device is if the ratio of current output to current input is 1.6.

$$\text{Efficiency} = \frac{\text{power output}}{\text{power input}} \text{ x } 100\%$$

$$= \frac{I_s V_s}{I_p V_p} \text{ x } 100\%$$

$$= \frac{1.6 \text{ x } 120}{240}$$

$$= 80\%$$

The transformer is 80% efficient

IR - 03 (10-14)
- 02 (5-9)
- 01 (1-4)
- 0 (0)

(d) Advise the welders on how to improve the efficiency of the device.

- By using soft iron core
- By using a laminated core
- By winding the secondary coil on top of primary coil
- By using a coil made of thick copper wires of low resistance.

TOTAL MAXIMUM SCORE = 6 SCORES

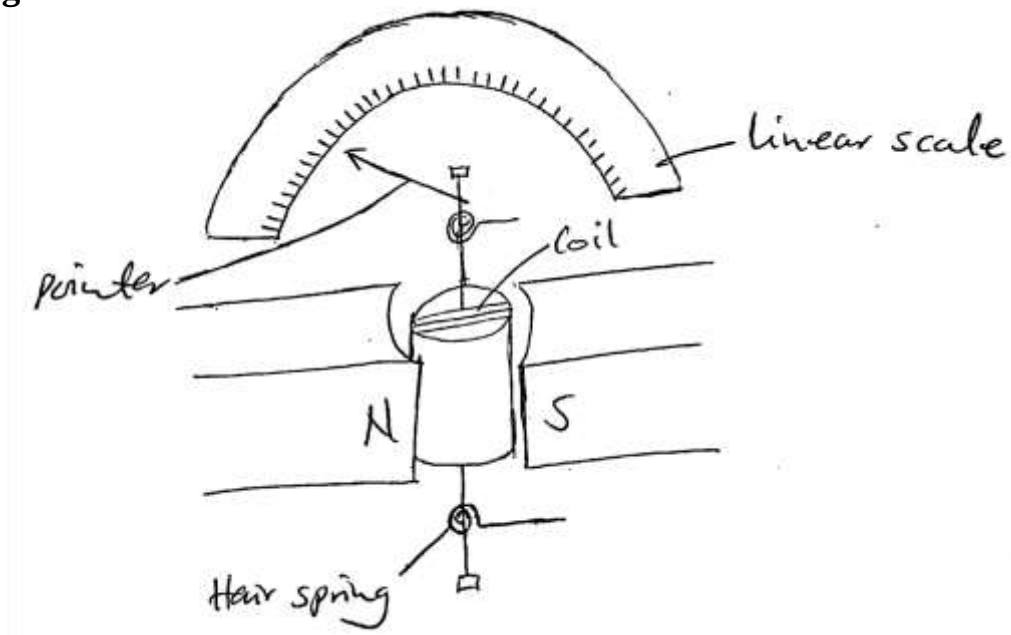
ITEM 7

In one of the secondary schools in a remote village, the students lack enough physics apparatus. The government decided to donate apparatus, including many moving coil galvanometers through a local supplier. However, there was still a scarcity of ammeters in the school. Each galvanometer has a resistance of 4 ohms and a full-scale deflection of 25 mA. The Head teacher wants to return some of the galvanometers back to the

supplier in exchange for ammeters. The Head teacher does not know the use and mode of operation of these galvanometers. However, the Physics teacher advises the Head teacher that these galvanometers can be modified to work as ammeters and can read up to a current of 1A. He added that this project can be entrusted to the senior four class where you are the physics class coordinator.

Task

Advise the Head teacher on how you are going to perform this task and explain to him how the galvanometers work.



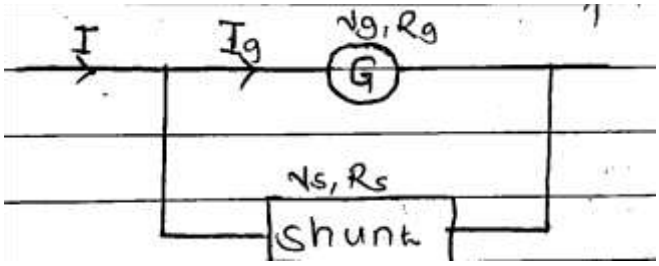
N,S are pole pieces of a permanent magnet

S_1 & S_2 are fine hair springs

I_1 & I_2 are jerolled bearings

When current to be measured is passed into coil through the hair spring S_1, the coil experiences a deflecting torque due to a couple of forces. The coil then turns in the magnetic field together with the pointer until it is stopped by the rotating torque in the hair springs. The pointer deflects over the linear scale.

To convert the moving coil galvanometer into an ammeter, the circuit is connected as shown below.



$I_S = I - I_y$

$= 1 - 25 \times 10^{-3}$

$= 0.975A$

$V = I_g R_g$

$= 25 \times 10^{-3} \times 4$

$= 0.1V$

$R_s = \frac{v}{I_s}$

$= \frac{0.1}{0.975}$

$= 0.1026\Omega$

To connect a moving coil galvanometer into an ammeter, a resistor of low resistance 0.1026Ω known as shunt has to be connected in parallel with the coil in the moving coil galvanometer.

MC_5 - 03 (9-12)
- 02 (3-8)
- 01 (1-2)
- 0 (0)

IR_5 - 03 (7-10)
- 02 (3-6)
- 01 (1-2)
- 0 (0)

TOTAL MAXIMUM SCORE = 6 SCORES

FINAL TOTAL MAXIMUM SCORE = 24

END